Progress Update

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July 2026

Summary

During my first week at the MPI-IS, I familiarised myself with the CAPT motor platform, the dSPACE control architecture, and the Simulink workflow. Alongside the onboarding process, I began contributing to the experimental software infrastructure while exploring several control strategies for future implementation.

Additionally, I assisted in the demo of the motor at the Open House, which was a very interesting event at the intersection of technology and the MPI community.

Technical Progress

I studied the existing virtual environments, including asymmetric springs, position control and configurable stiffness parameters, while investigating the effects of virtual damping and inertia on the motor's haptic behaviour. I also explored impedance estimation using motor torque and encoder position measurements.

A major focus was the development of a Python-based GUI using `PyQt6`. The current implementation includes a modular tab-based interface with real-time plotting, history recording and initial stability visualisation, providing a foundation for future impedance, passivity and transparency analysis.

To connect the real-time controller with the GUI, I implemented both a UDP communication interface and a shared-memory framework between Simulink and Python. These enable low-latency transfer of experimental data and provide a flexible architecture for real-time monitoring and controller evaluation.

From the control perspective, I designed an impedance-inspired inertia compensation controller to improve the transparency of the motor while maintaining stability (tried with a physical model of the motor, to be tested on the motor). I also incorporated model linearisation and a Padé approximation to compensate for Zero-Order Hold effects, together with stability visualisation integrated into the GUI.

Objectives for next Week

The next steps are to further develop the GUI, refine the impedance-inspired controller, and begin implementing and evaluating energy-based and passivity-preserving control strategies on the experimental platform.